

Memory

Parallel and Heterogeneous Computing

David Mayerich

Scalable Tissue Imaging and Modeling (STIM) Laboratory

Department of Electrical and Computer Engineering

Cullen College of Engineering

University of Houston

UNIVERSITY of **HOUSTON** | ECE

Processor Instructions

- **cycle time**: length (in seconds) for a clock cycle:
 $5\text{GHz} = 2 \times 10^{-10}$ seconds (≈ 10 ns)
- **step**: execution of an instruction
- **step-rate**: number of instructions executed in a cycle
- **cycles per instruction (CPI)**: number of clock cycles to execute the instruction

- Intel Tiger Lake:

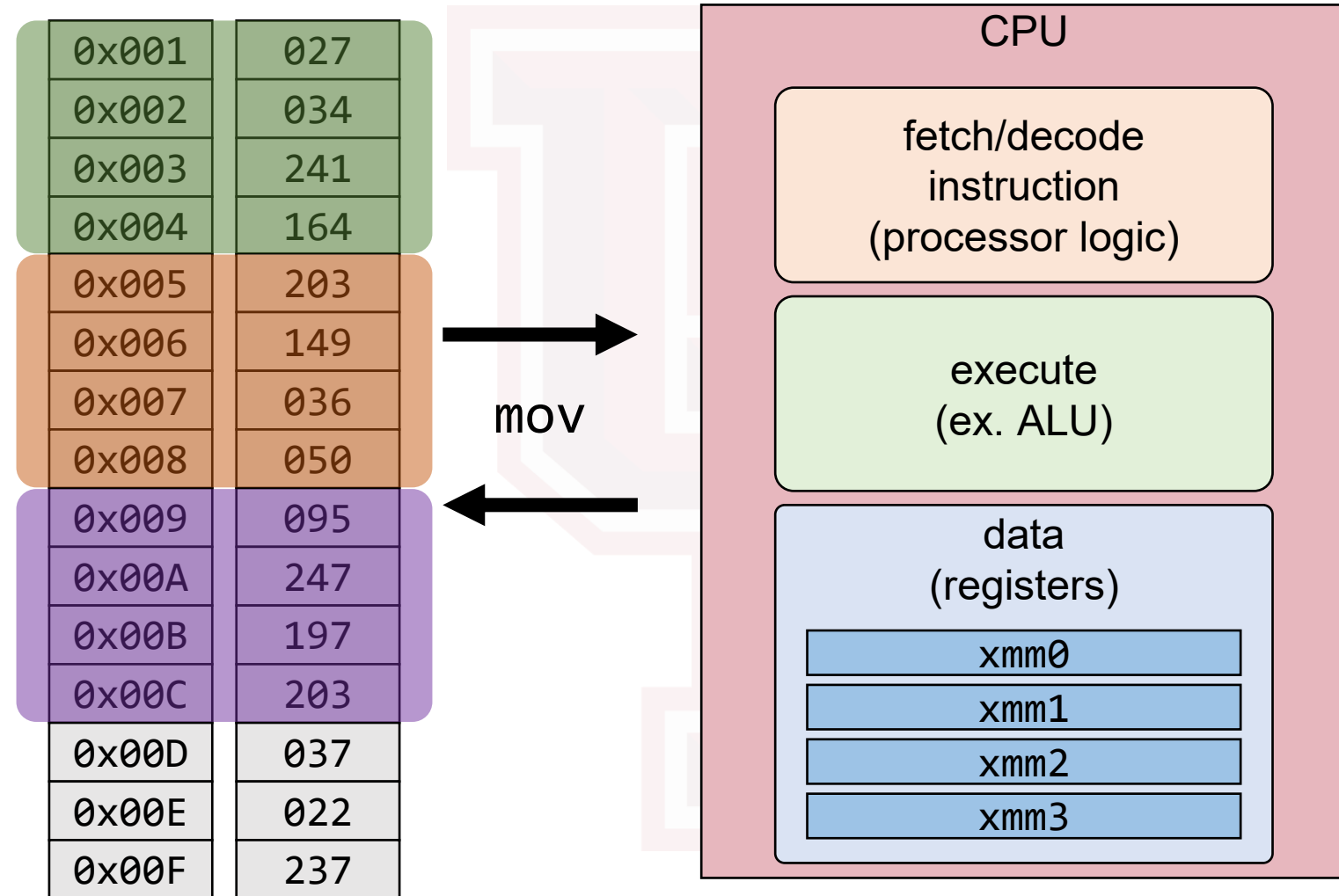
Instruction	CPI
ADD, SUB	0.25
MUL	1.00
DIV	6.00
FADD	1.00
FMUL	1.00
FDIV	4.00
• Memory Latency:	
DDR	≈ 300

Memory Instructions

- Memory is a 1D array of values
- Load/Store instructions move data between memory and registers

```
float a = x*x + y*y + z*z;
```

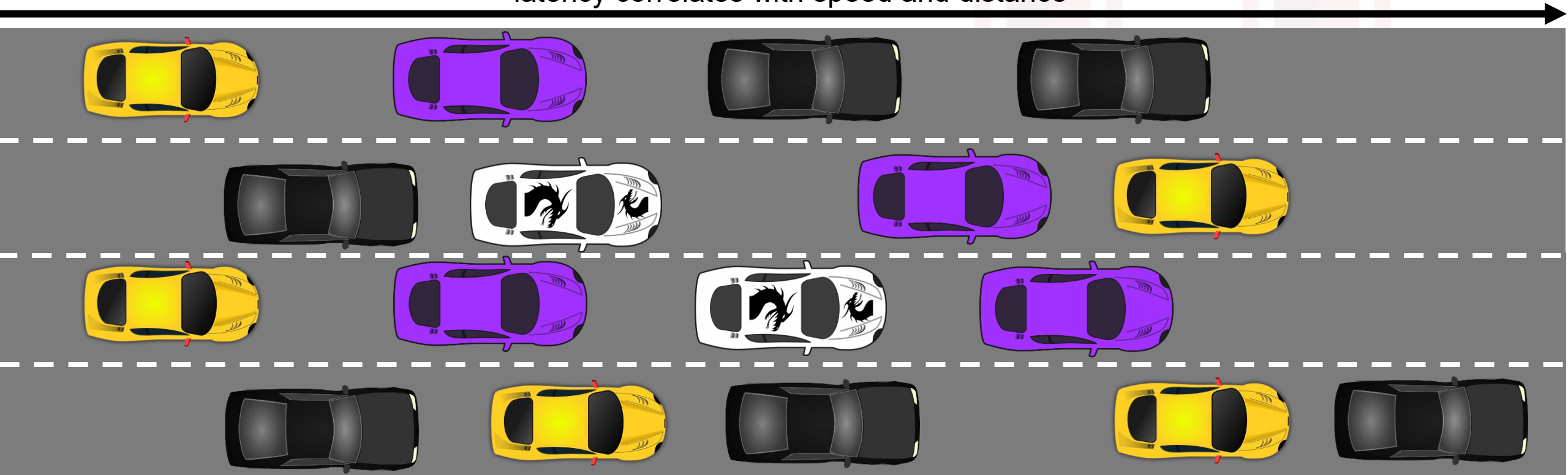
```
mov    xmm1, DWORD PTR x[rip]
mov    xmm0, DWORD PTR x[rip]
mul    xmm1, xmm0
mov    xmm2, DWORD PTR y[rip]
mov    xmm0, DWORD PTR y[rip]
mul    xmm0, xmm2
add    xmm1, xmm0
mov    xmm2, DWORD PTR z[rip]
mov    xmm0, DWORD PTR z[rip]
mul    xmm0, xmm2
add    xmm0, xmm1
mov    DWORD PTR a[rip], xmm0
```



Latency vs. Bandwidth

- latency: the time it takes for a single piece of data to be retrieved
- bandwidth: the amount of data that can be transferred in a time span

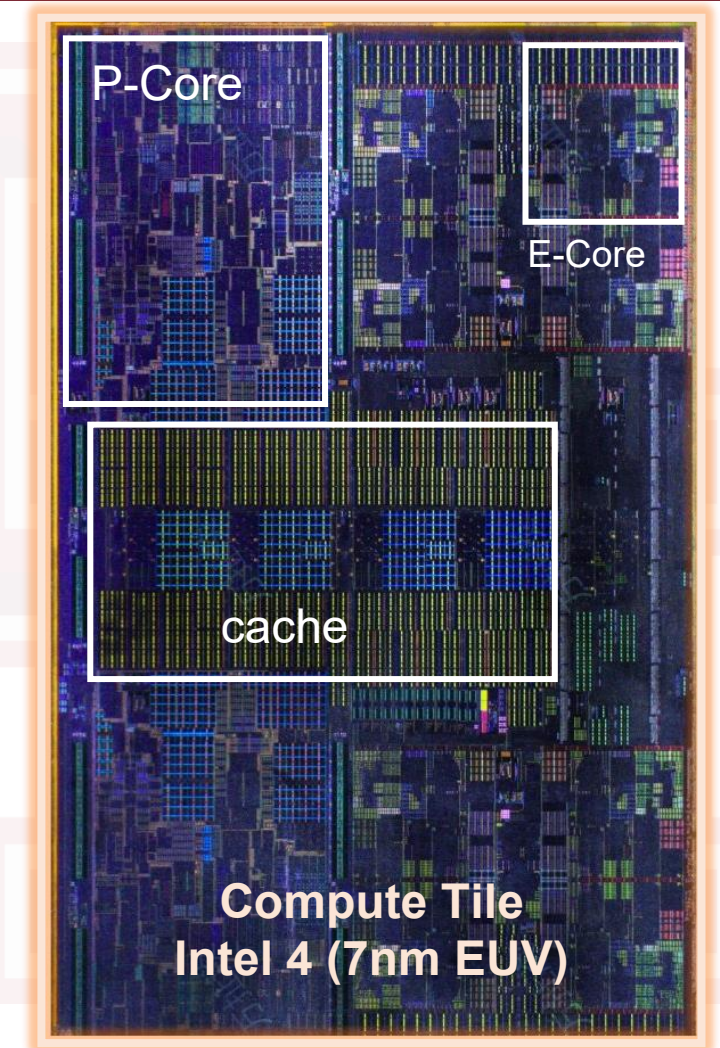
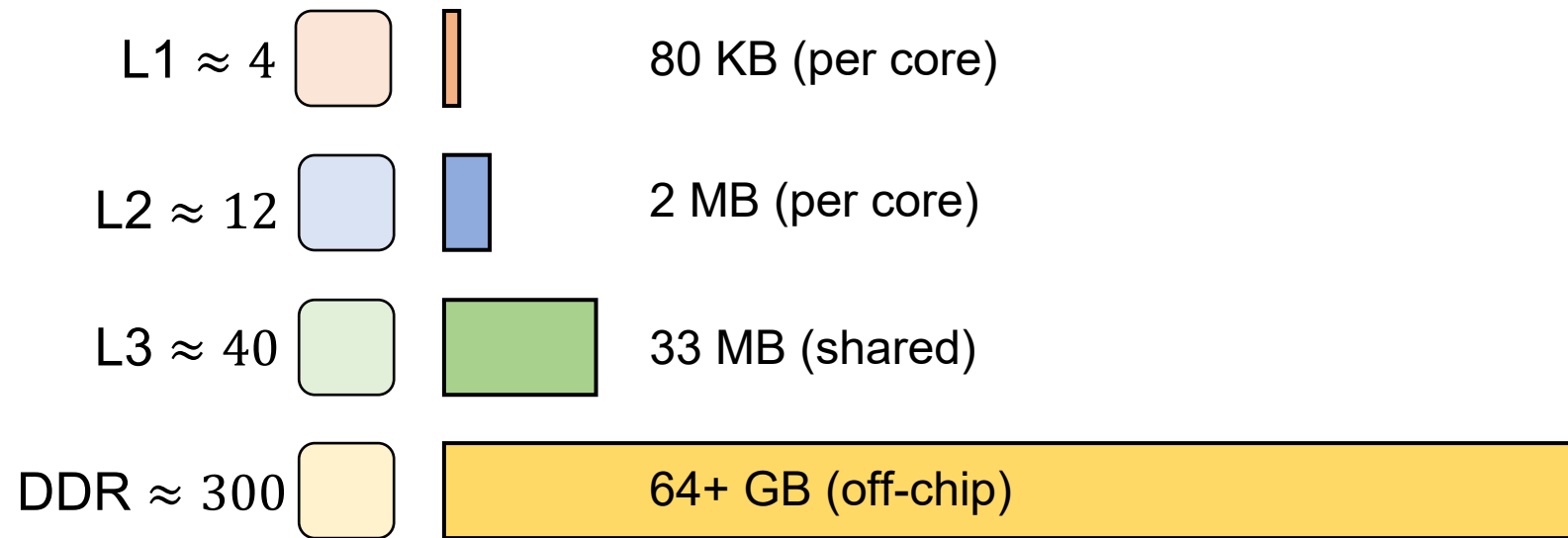
latency correlates with speed and distance



Caches

- Limited storage, close to the processor
- Multiple stages with increasing speed

Core i7-14700F (P-Cores)



Linear Array Access

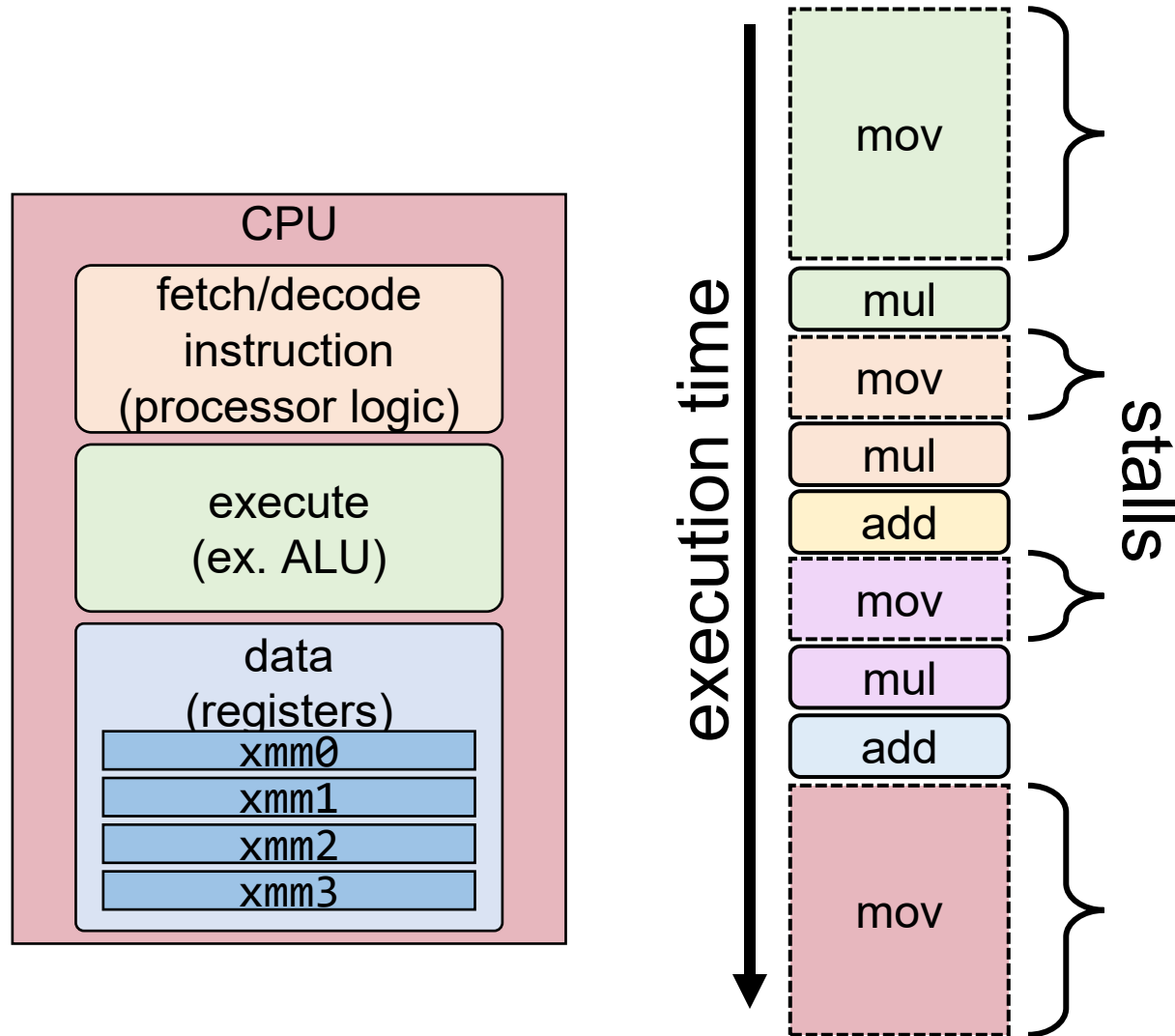
0x001	027
0x002	034
0x003	241
0x004	164
0x005	203
0x006	149
0x007	036
0x008	050
0x009	095
0x00A	247
0x00B	197
0x00C	203
0x00D	037
0x00E	022
0x00F	237

4-byte
cache
lines

```
mov    xmm1    0x001    miss, load 0x001
mov    xmm1    0x002    hit
mov    xmm1    0x003    hit
mov    xmm1    0x004    hit
mov    xmm1    0x005    miss, load 0x005
mov    xmm1    0x006    hit
mov    xmm1    0x007    hit
mov    xmm1    0x008    hit
mov    xmm1    0x009    miss, load 0x009
mov    xmm1    0x00A    hit
mov    xmm1    0x00B    hit
mov    xmm1    0x00C    hit
mov    xmm1    0x00D    miss, load 0x00D
```

cache	
0x009	095
0x00A	247
0x00B	197
0x00C	203
0x00D	037
0x00E	022
0x00F	237
0x008	050

Execution Stalls



- Memory reads cause stalls in fetch/decode and execution nodes

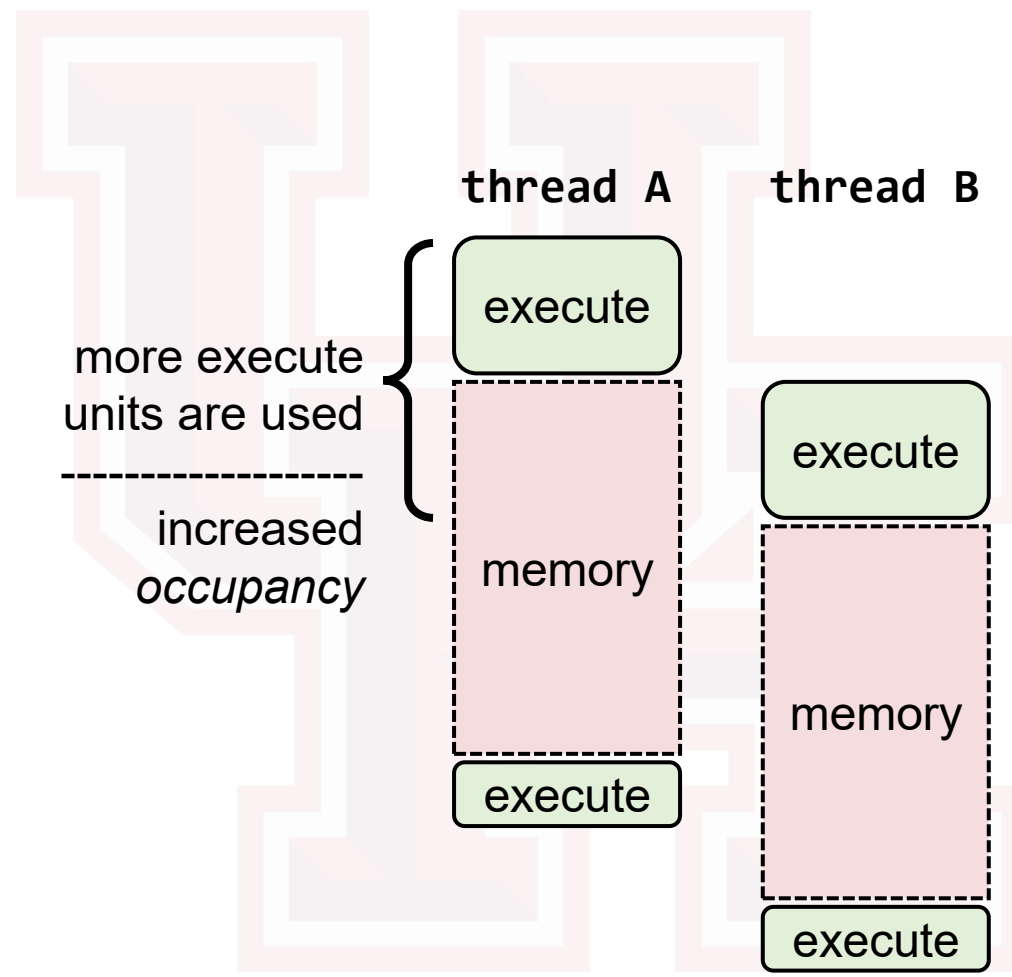
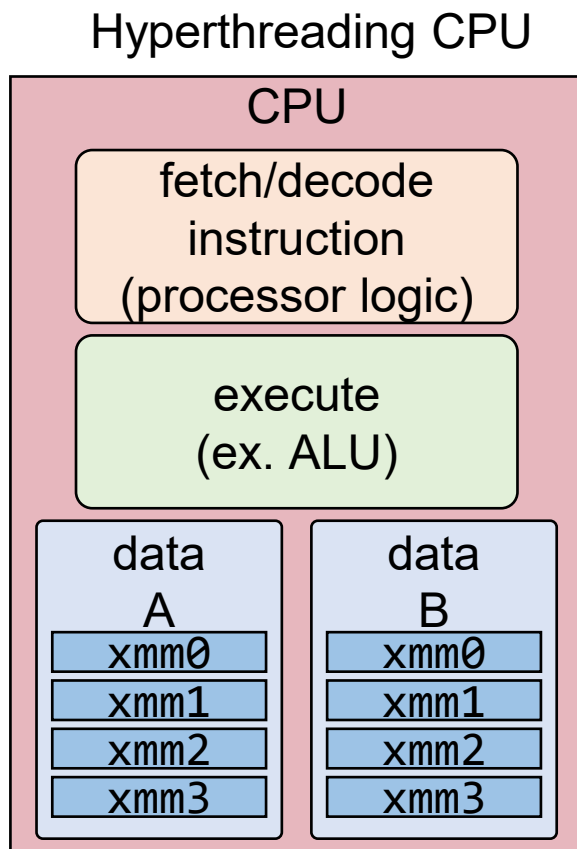
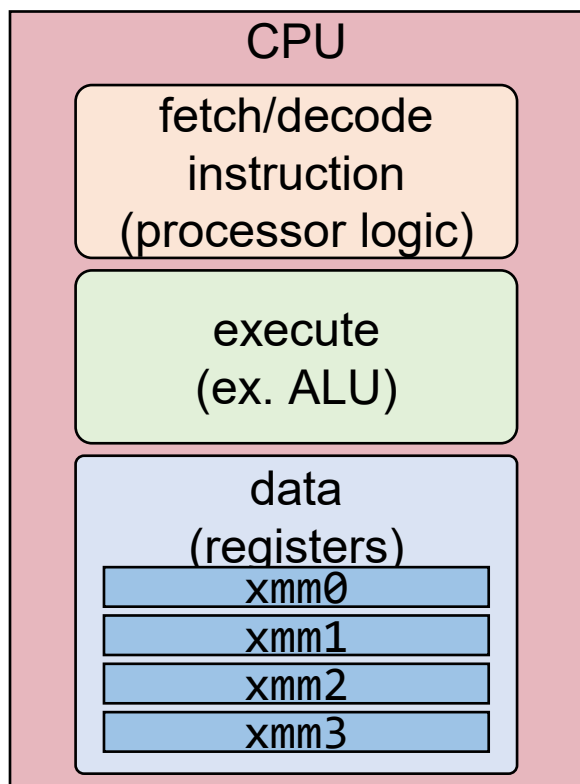
float **a** = **x*****x** + **y*****y** + **z*****z**;

```

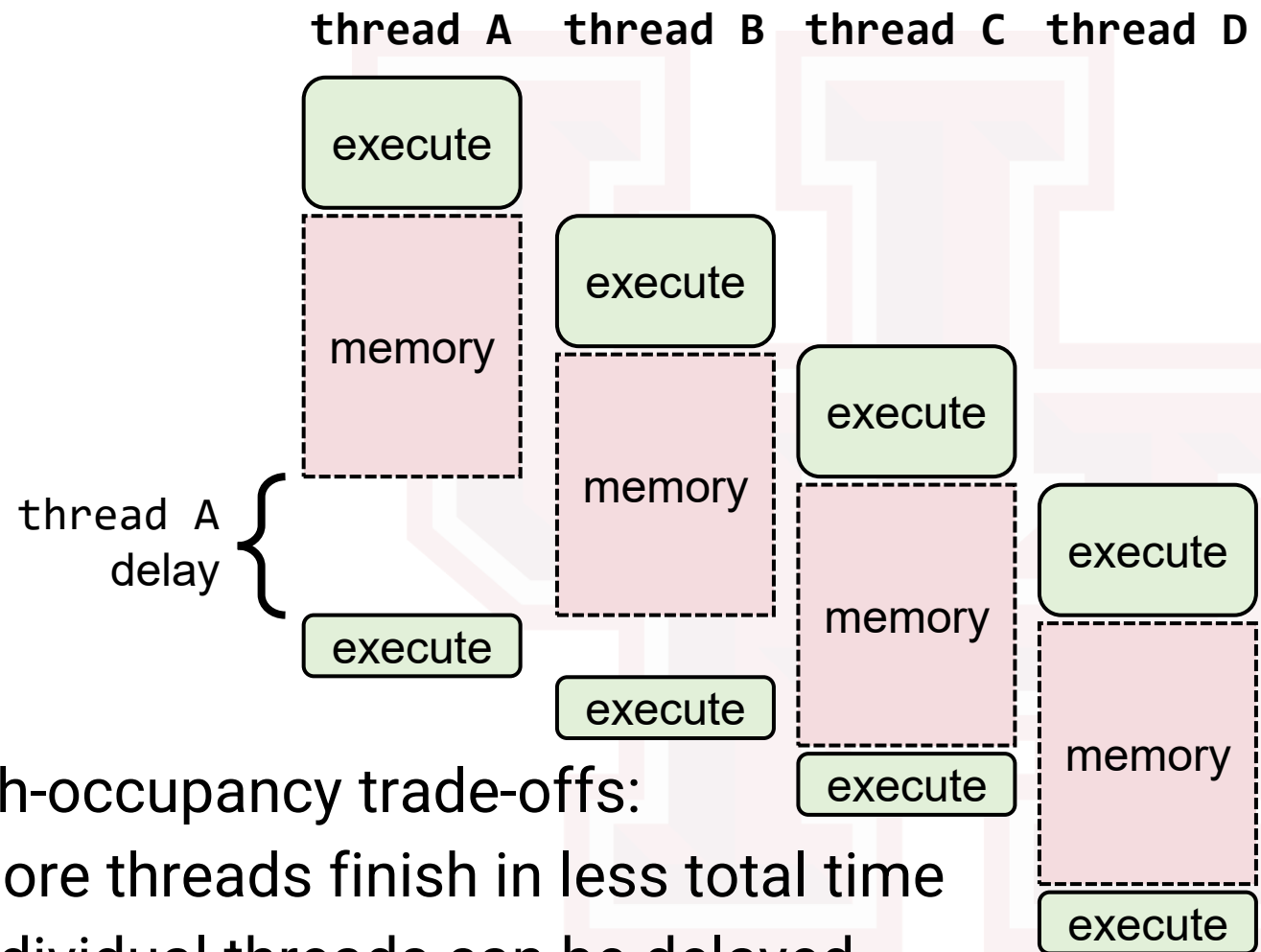
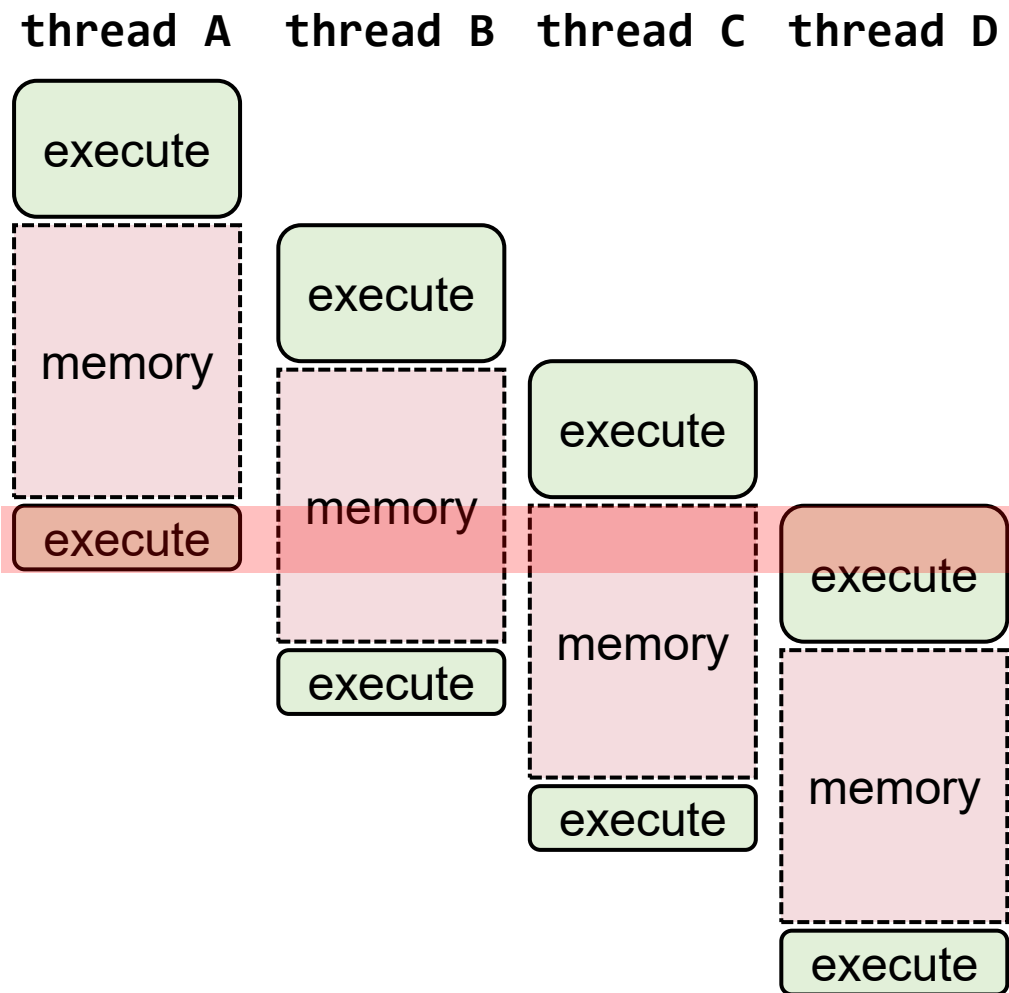
mov    xmm1, DWORD PTR x[rip]
mov    xmm0, DWORD PTR x[rip]
mul    xmm1, xmm0
mov    xmm2, DWORD PTR y[rip]
mov    xmm0, DWORD PTR y[rip]
mul    xmm0, xmm2
add    xmm1, xmm0
mov    xmm2, DWORD PTR z[rip]
mov    xmm0, DWORD PTR z[rip]
mul    xmm0, xmm2
add    xmm0, xmm1
mov    DWORD PTR a[rip], xmm0
  
```


Hyperthreading

- “Hide” stalls to keep execute nodes active



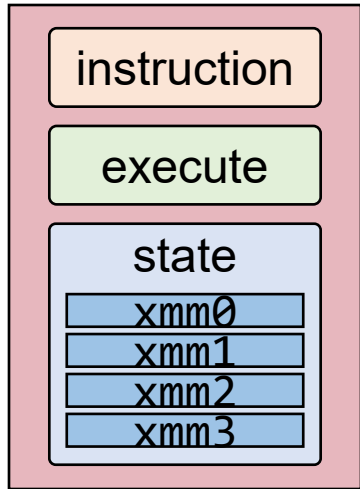
Maximizing occupancy can delay threads



High-occupancy trade-offs:

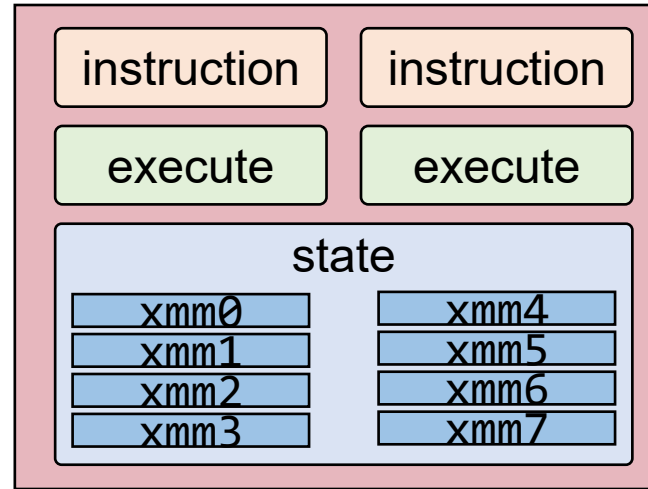
- more threads finish in less total time
- individual threads can be delayed

Processor Types (So Far)



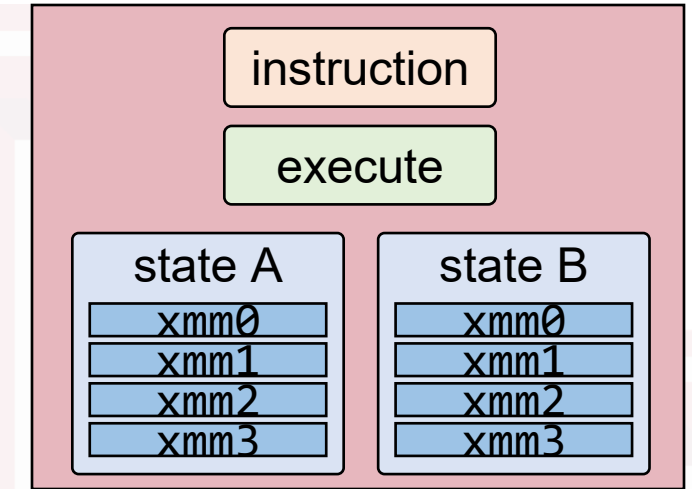
Single-Core (SISD)

- One instruction
- One operation
- One state space



Superscalar Processor

- Instruction-level parallelism
- All ALUs have the same state space

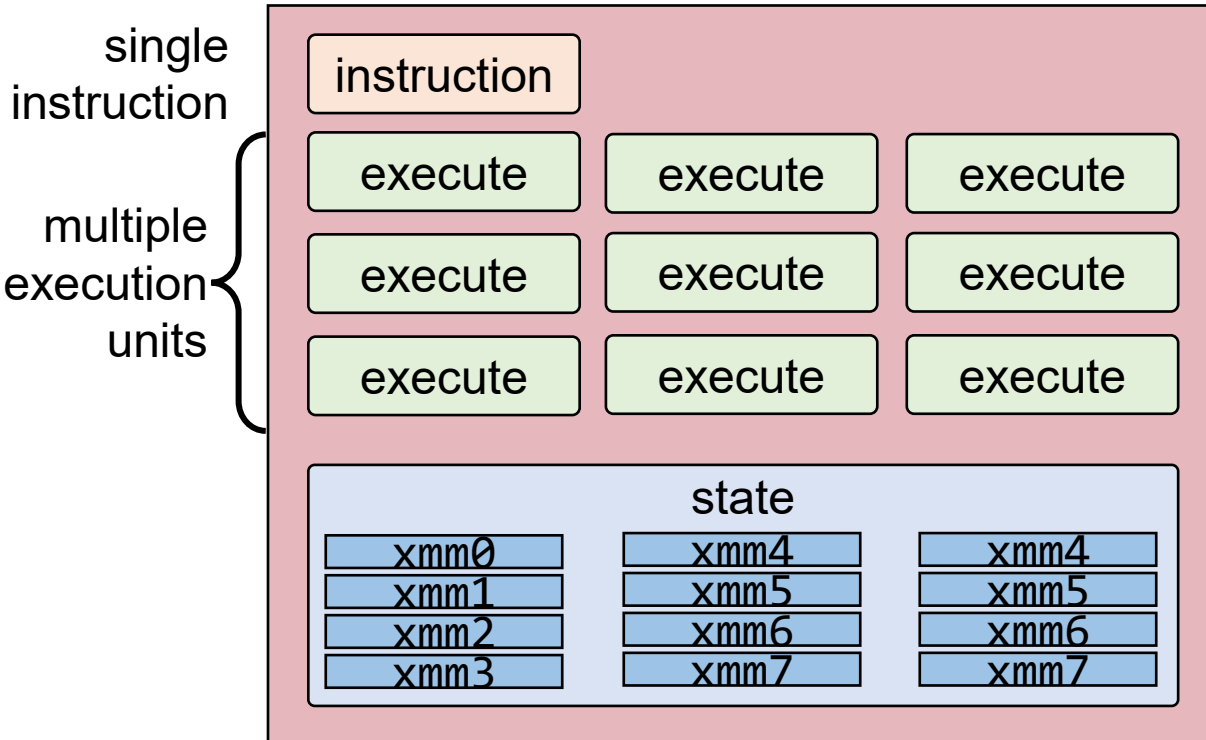


Hyperthreading Processor

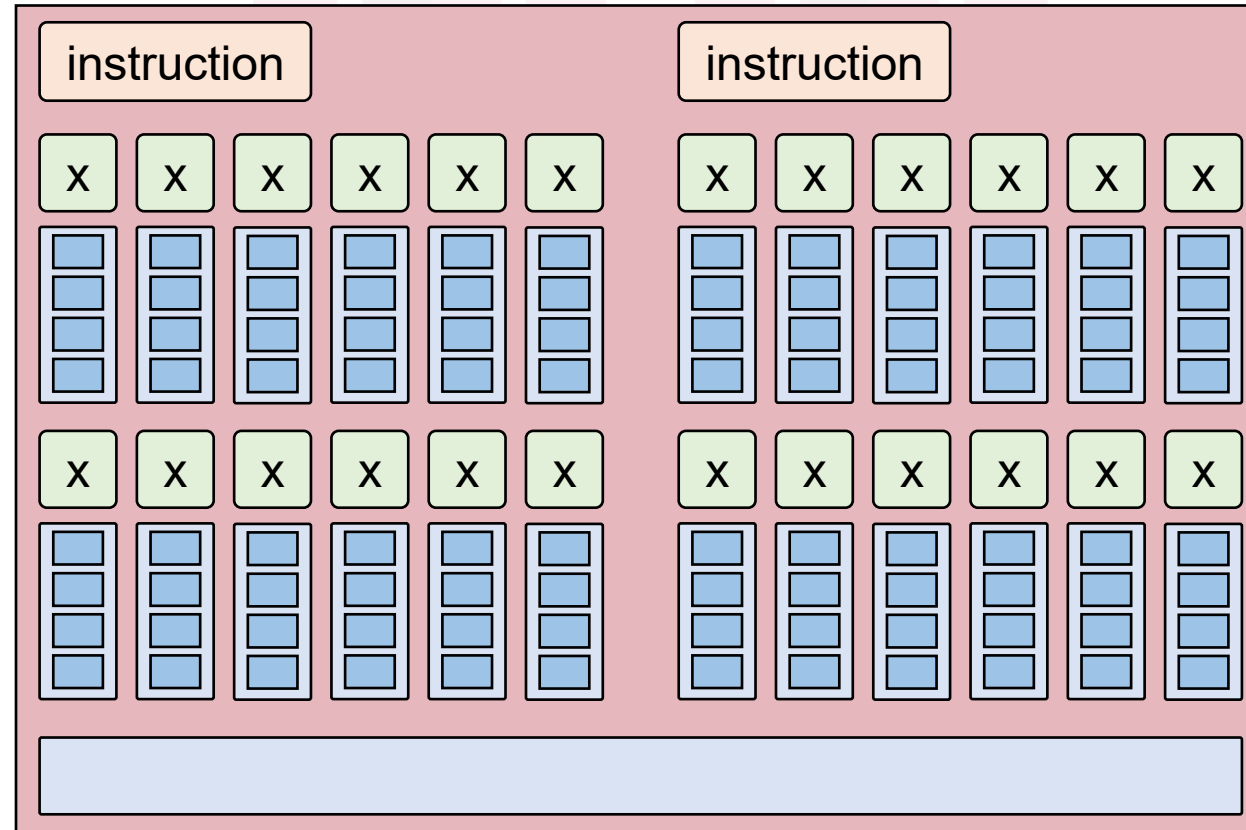
- Single instruction
- Independent state spaces
- Interspersed instructions prevent stalls

Identify the Application

Vector Processor (SIMD)

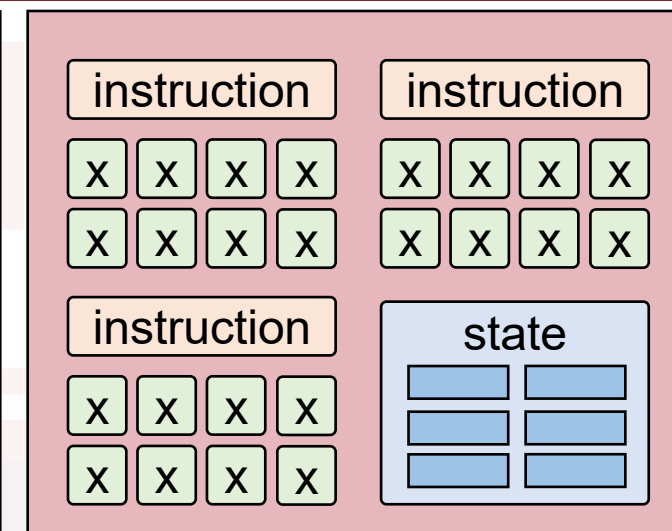
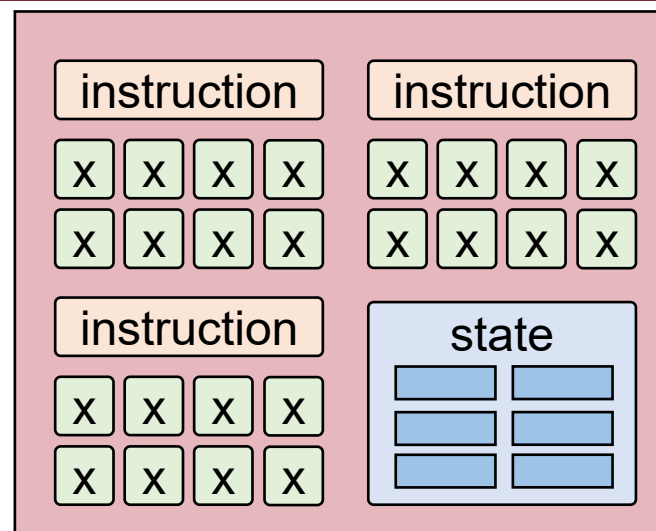
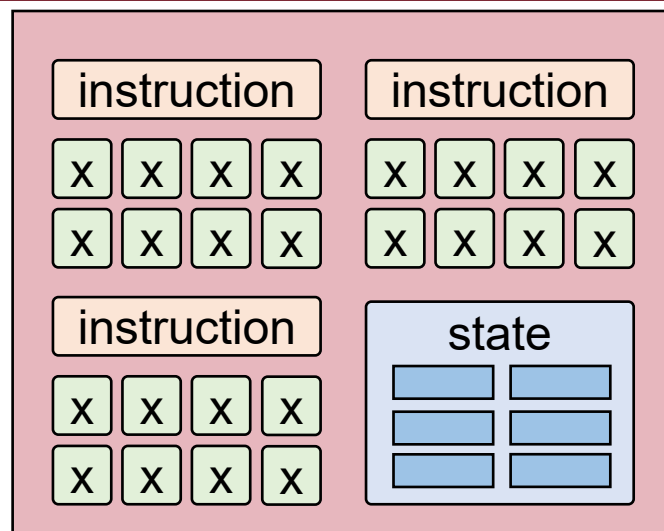


Graphics Processor



Intel Kaby Lake (i7-7700K)

- 4-core processor
- 3 8-unit SIMD
- Intel AVX-512



- Throughput: 500 GFLOPs: $4 \times 8 \times 3 \times 4.2\text{GHz}$
- Compiler support: GCC, VS (/arch:AVX), Intel oneAPI
- Intel ISPC compiler: <https://ispc.github.io/>
- <https://pharr.org/matt/blog/2018/04/30/ispc-all>

